



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

EXPERIMENT NO – 9

1. **AIM** - Implement localStorage and sessionStorage to save user preferences like theme settings.
2. **SOFTWARE REQUIRED** - Visual Studio Code, Web Browser (Google Chrome), HTML, CSS, JavaScript, Live Server extension.

3. THEORY –

Web storage APIs allow web applications to store data locally within the user's browser securely. Before HTML5, application data had to be stored in cookies, included in every server request. Web storage is more secure, and large amounts of data can be stored locally without affecting website performance.

There are two mechanisms for local storage in the browser:

- **LocalStorage:**
 - Stores data with **no expiration date**.
 - The data remains saved in the browser even after the browser window or tab is closed and reopened.
 - Ideal for saving long-term preferences like user themes or language settings.
- **SessionStorage:**
 - Stores data only for the duration of the current page session.
 - The data is **completely wiped out** as soon as the user closes that specific browser tab or window.
 - Ideal for sensitive temporary data or session-specific states.

Both storage types store data in simple **key-value pairs**, and the data is always stored as strings.



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

4. OUTPUT –

The screenshot shows a web browser window with the address bar displaying 'C:/Users/aryak/Downloads/JS%20exp%209/index.html'. The main content area is a white box with a purple border. It contains a 'Choose a Theme' section with four buttons: 'Light' (selected), 'Dark', 'Pink', and 'Green'. Below this is a 'Select Storage Type' section with two radio buttons: 'Local Storage' (selected) and 'Session Storage'. A 'Submit' button is located below the storage options. At the bottom of the box, it says 'Current Storage Status: Saved "light" to Local Storage.' followed by a horizontal line. Below the line, it says 'Developed by name: Arya Khandekar' and 'PRN: 24070521219'.

5. CONCLUSION –

In this practical, the implementation of both LocalStorage and SessionStorage was successfully demonstrated by saving and retaining custom UI theme configurations (Light, Dark, Pink, and Green). It can be concluded that LocalStorage is highly effective for maintaining persistent user preferences across browser sessions, whereas SessionStorage serves as an efficient tool for maintaining temporary, session-isolated browser preferences.